Refining the recipe or spoiling the broth? Reframing perceptions of programme adaptation in sports injury prevention

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Multiple injury prevention exercise programmes have demonstrated efficacy under randomised controlled conditions.[1] However, successfully implementing these programmes under real-world conditions poses a significant challenge. It is well documented that football teams modify programmes to better fit their specific context, drawing not only on evidence from randomised controlled trials (RCTs) but also their past experiences, guidelines and individual player screening data.[2] Furthermore, evaluation of injury prevention programmes under real-world conditions demonstrates that increased programme adaptability, variation and individualisation can facilitate implementation efforts.[2] Modifying efficacious programmes can potentially enhance context-specificity but may also jeopardise fidelity. Replicating the exact protocol from an RCT increases the chances of achieving the desired injury reduction, but failing to consider the intricacies of the specific implementation context and modify the intervention accordingly increases the risk of poor adherence. Balancing these dual aspirations of fidelity and context-specific modifications has been coined the adaptation-fidelity dilemma.[3]

RETHINKING ADHERENCE

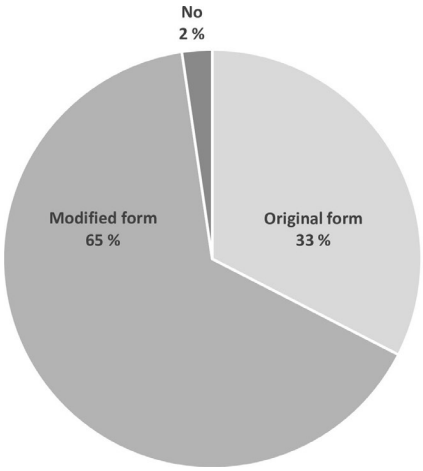
To evaluate implementation outcomes, sports injury prevention researchers have embraced the Reach Effectiveness Adoption Implementation Maintenance (RE-AIM) framework.[4–6] While this has heightened awareness of different implementation factors and common pitfalls, most analyses only consider teams' use of interventions in their original (RCT) form. But programme modifications are common[2 7] and failing to assess and analyse them will paint an incomplete picture of real-world adherence. Taking the example of the Copenhagen Adduction Exercise (CAE), under RCT conditions, the programme reduced the risk of groin problems (41%)[8] and scored highly for the RE-AIM domains of Reach (100%),[7] Adoption (100%)[7] and Maintenance (68%–97%).[5 7] However, there were significant challenges with fidelity (often termed 'adherence' and termed 'implementation' in RE-AIM), which describes the extent to which an intervention is performed as originally intended.[4] Only 46%[5] of players and 10%[7] of the staff delivering the programme (eg, physiotherapists) reported adhering to the original protocol.

In a recent cross-sectional study,[9] we asked 50 European football teams about their use of the CAE (figure 1). Of the 42 teams that knew the CAE, one-third (33%) adhered to the original (RCT) protocol and two-thirds (65%) reported using modified forms of the CAE programme. These modifications still observed the core components[10]: a progressive hip adduction strengthening programme. When adherence is defined as performing the core components of the programme, in original or modified form, the adherence rate jumps to 98%.

Strict adherence to an RCT protocol is analogous to strictly adhering to a cooking recipe[3]: by replicating the ingredients and carefully following sequential steps, the desired outcome is achieved, for example, a tasty lasagna. However, lasagna comes in many variations and sizes and modifying the recipe can have both negative and positive impacts. Furthermore, one needs to cook for the occasion by selecting the right recipe and amounts for the specific circumstance. This requires a comprehensive cookbook with multiple, adaptable recipes. The key question is, which modifications can be made without spoiling the outcome. Great recipes will evolve over time, with cooks employing a hybrid approach[3] of maintaining certain key ingredients and processes but carefully manipulating others over multiple iterations. In sports medicine, the traditional view of implementation has encouraged teams to exactly replicate a single recipe (RCT protocol), indefinitely and irrespective of contextual needs and situations. Any deviation from the path is perceived as synonymous with a negative impact on the programme's fidelity.[10]

MOVING BEYOND ONE-SIZE-FITS-ALL

We believe it is high time to move beyond the one-size-fits-all approach to injury prevention and embrace contextual modifications that respect key ingredients, along with the use of multiple programmes to cater for different occasions and athletes. We believe the focus should be on identifying and evaluating the key ingredients (eg, strength) and processes (eg, achieving a certain intervention dose).[11] Teams can then employ a 'hybrid approach',[3]



Adherence	
RCT-Recipe	33%
Key components	98%

Figure 1 Use of the Copenhagen Adduction Exercise by 50 European football teams (Dalen-Lorentsen).[9] RCT, randomised controlled trial.

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with certain elements remaining constant and others being modified to the unique context. Professional teams in particular, with their extensive staff and facilities, can be expected to modify programmes substantially. Importantly, this does not condone careless, spontaneous and unilateral modifications, similar to randomly omitting key recipe ingredients due to laziness or poor preparation.[3] Rather, it promotes a thoughtful, planned process, engaging all relevant staff members, players and potentially researchers. This ensures that essential ingredients are retained, while others are modified to increase contextual fit. Adding 'spice' in the form of variation, individualisation and fun elements will further enhance the fit to specific athletes and settings and enhance adherence. Ongoing evaluation of key outcomes (eg, injury rates, strength values and adherence levels) can drive the process and ensure that modifications do not go too far.[11]

As an example of purposeful and considered modification, teams in the aforementioned study[9] reported modifying the level 1 CAE. Rather than performing side-lying hip adduction, whereby the lower leg is raised from the floor, teams employed an isometric adduction exercise in a side plank position, with the player's upper knee and ankle supported on a training bench. Staff believed the modified exercise prepared players better for the level 2 exercise, with the additional benefit of simultaneously training the hip adductors and trunk muscles. This made the exercise a more time-efficient option for staff members juggling multiple training goals and priorities.[2 9] Additionally, when progressing to level 2, the use of a bench eliminated the need for a partner, meaning that all players could be loaded simultaneously in a gym-based circuit session.

In conclusion, a paradigm shift is needed that embraces considered, purposeful modification to efficacious interventions while respecting the key ingredients and processes. From a research perspective, this will require moving beyond the view of implementation as a one-way, linear process spanning from RCT to real-world practice. Instead, the development of preventive programmes should be considered a dynamic, iterative process and product of both researcher and practitioner contributions. Researchers must strive to identify and assess core intervention components[10] (ingredients) and core implementation components[10] (processes) to help inform practitioners' modifications. Furthermore, researchers must heed previous calls to define, assess and report adherence outcomes consistently.[12] Practitioners working at the injury prevention coalface can contribute by carefully evaluating and documenting the what, how, why and by whom of exercise modifications. Along with helping to avoid spontaneous and poorly considered modifications, this information can also fuel the development of future programmes and the scientific analysis of the impact of modifications on fidelity. The ultimate beneficiary will be the athlete with reduced injury risk.

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